

Solutions to Exercise Sheet 2: Groups and Lie Algebras

Exercise 1: Simple groups

Definition. A group G is called *simple* if its only normal subgroups are the trivial subgroup $\{1\}$ and the whole group G .

Definition. If $\varphi : G \rightarrow H$ is a group homomorphism, its *kernel* is

$$\ker(\varphi) = \{g \in G : \varphi(g) = 1_H\}.$$

The kernel is always a normal subgroup of G .

Definition. A map $f : G \rightarrow H$ is called *injective* if

$$f(g_1) = f(g_2) \implies g_1 = g_2.$$

A map is called *constant at the identity* if it sends every element of G to 1_H .

1(a)

Let G be simple and let $\varphi : G \rightarrow H$ be a group homomorphism. Since $\ker(\varphi)$ is a normal subgroup of G , and G is simple, there are only two possibilities:

$$\ker(\varphi) = \{1\} \quad \text{or} \quad \ker(\varphi) = G.$$

If $\ker(\varphi) = \{1\}$, then φ is injective. Indeed, suppose

$$\varphi(g_1) = \varphi(g_2).$$

Then

$$\varphi(g_1 g_2^{-1}) = \varphi(g_1) \varphi(g_2)^{-1} = 1_H,$$

so $g_1 g_2^{-1} \in \ker(\varphi) = \{1\}$. Hence $g_1 = g_2$.

If $\ker(\varphi) = G$, then every element of G is mapped to the identity of H , so φ is the constant map at the identity.

Thus every homomorphism from a simple group to any group is either injective or constant at the identity.

Conversely, suppose G' is a group with the property that every group homomorphism from G' to any group is either injective or constant at the identity. We show that G' is simple.

Let $N \trianglelefteq G'$ be a normal subgroup. Consider the quotient map

$$\pi : G' \rightarrow G'/N, \quad \pi(g) = gN.$$

This is a group homomorphism. By assumption, π is either injective or constant at the identity.

If π is injective, then its kernel is trivial. But

$$\ker(\pi) = N,$$

so $N = \{1\}$.

If π is constant at the identity, then

$$\pi(g) = N \quad \text{for all } g \in G'.$$

This means every $g \in G'$ lies in N , so $N = G'$.

Hence the only normal subgroups of G' are $\{1\}$ and G' itself. Therefore G' is simple.

1(b)

We determine for which $n \in \mathbb{N}$ the group $\mathbb{Z}/n\mathbb{Z}$ is simple.

Since $\mathbb{Z}/n\mathbb{Z}$ is abelian, every subgroup is automatically normal. Therefore $\mathbb{Z}/n\mathbb{Z}$ is simple if and only if it has no nontrivial proper subgroups.

If n is composite, say $n = ab$ with $1 < a < n$, then the subgroup generated by $[a]$ is a nontrivial proper subgroup. So $\mathbb{Z}/n\mathbb{Z}$ is not simple.

If $n = p$ is prime, then every nonzero element generates the whole group, so the only subgroups are

$$\{[0]\} \quad \text{and} \quad \mathbb{Z}/p\mathbb{Z}.$$

Hence $\mathbb{Z}/p\mathbb{Z}$ is simple.

Therefore,

$$\mathbb{Z}/n\mathbb{Z} \text{ is simple if and only if } n \text{ is prime.}$$

1(c)

Let G be a group and let $N \trianglelefteq G$ be a maximal normal subgroup, meaning that

$$N \neq G,$$

and whenever $N \subsetneq N' \subseteq G$ with $N' \trianglelefteq G$, then $N' = G$.

We show that G/N is simple.

Let

$$K \trianglelefteq G/N$$

be a normal subgroup. Define

$$M = \{g \in G : gN \in K\}.$$

We will show that M is a normal subgroup of G containing N .

First, M is a subgroup. Indeed, if $g, h \in M$, then $gN, hN \in K$. Since K is a subgroup of G/N , we have

$$(gN)(hN)^{-1} \in K.$$

But

$$(gN)(hN)^{-1} = (gN)(h^{-1}N) = gh^{-1}N,$$

so $gh^{-1}N \in K$, hence $gh^{-1} \in M$. Therefore M is a subgroup of G .

Next, $N \subseteq M$. If $n \in N$, then

$$nN = N,$$

and N is the identity element of G/N , so $N \in K$. Hence $n \in M$.

Now we show that M is normal in G . Let $g \in G$ and $m \in M$. Since $mN \in K$ and K is normal in G/N , we have

$$(gN)(mN)(gN)^{-1} \in K.$$

Computing this gives

$$(gN)(mN)(gN)^{-1} = gmg^{-1}N,$$

so $gmg^{-1}N \in K$, hence $gmg^{-1} \in M$. Therefore $M \trianglelefteq G$.

Thus M is a normal subgroup of G containing N . Since N is maximal normal, it follows that either

$$M = N \quad \text{or} \quad M = G.$$

If $M = N$, then the only coset belonging to K is N , so

$$K = \{N\}.$$

If $M = G$, then every coset gN belongs to K , so

$$K = G/N.$$

Hence the only normal subgroups of G/N are $\{N\}$ and G/N itself. Therefore G/N is simple.

1(d)

We show that a group with only two conjugacy classes is simple.

Definition. The *conjugacy class* of an element $g \in G$ is

$$\mathcal{C}(g) = \{xgx^{-1} : x \in G\}.$$

Lemma. If $N \trianglelefteq G$ is a normal subgroup, then N is a union of conjugacy classes.

Proof. Let $n \in N$. We show that the whole conjugacy class of n is contained in N .

For any $g \in G$, since N is normal, we have

$$gNg^{-1} = N.$$

Because $n \in N$, it follows that

$$gng^{-1} \in N.$$

Thus every element conjugate to n lies in N , so

$$\mathcal{C}(n) \subseteq N.$$

Since this holds for every $n \in N$, the subgroup N is a union of conjugacy classes.

Suppose G has exactly two conjugacy classes. One of them is always

$$\{1\},$$

since the identity commutes with everything. Let the other conjugacy class be C . Then

$$G = \{1\} \sqcup C.$$

Let $N \trianglelefteq G$. Since N is a union of conjugacy classes and must contain the identity, there are only two possibilities:

$$N = \{1\} \quad \text{or} \quad N = \{1\} \cup C = G.$$

Thus G is simple.

1(e)

Definition. A *Jordan–Hölder decomposition* (composition series) of a finite group G is a chain

$$G = G_0 \triangleright G_1 \triangleright \cdots \triangleright G_r = \{1\}$$

in which each G_{i+1} is a maximal normal subgroup of G_i . The quotients G_i/G_{i+1} are then simple.

We carry this out for

$$G = \mathbb{Z}/12\mathbb{Z}.$$

Since $\mathbb{Z}/12\mathbb{Z}$ is abelian, every subgroup is normal. The maximal proper subgroups are

$$\langle [2] \rangle \cong \mathbb{Z}/6\mathbb{Z} \quad \text{and} \quad \langle [3] \rangle \cong \mathbb{Z}/4\mathbb{Z}.$$

First choice. Take

$$G_1 = \langle [2] \rangle.$$

Then

$$G/G_1 \cong \mathbb{Z}/2\mathbb{Z}.$$

Inside $G_1 \cong \mathbb{Z}/6\mathbb{Z}$, choose

$$G_2 = \langle [4] \rangle = \{[0], [4], [8]\} \cong \mathbb{Z}/3\mathbb{Z}.$$

Then

$$G_1/G_2 \cong \mathbb{Z}/2\mathbb{Z}, \quad G_2/\{0\} \cong \mathbb{Z}/3\mathbb{Z}.$$

So the simple factors are

$$\mathbb{Z}/2\mathbb{Z}, \quad \mathbb{Z}/2\mathbb{Z}, \quad \mathbb{Z}/3\mathbb{Z}.$$

Second choice. Take

$$G'_1 = \langle [3] \rangle.$$

Then

$$G/G'_1 \cong \mathbb{Z}/3\mathbb{Z}.$$

Inside $G'_1 \cong \mathbb{Z}/4\mathbb{Z}$, take

$$G'_2 = \langle [6] \rangle = \{[0], [6]\} \cong \mathbb{Z}/2\mathbb{Z}.$$

Then

$$G'_1/G'_2 \cong \mathbb{Z}/2\mathbb{Z}, \quad G'_2/\{0\} \cong \mathbb{Z}/2\mathbb{Z}.$$

So the simple factors are

$$\mathbb{Z}/3\mathbb{Z}, \quad \mathbb{Z}/2\mathbb{Z}, \quad \mathbb{Z}/2\mathbb{Z}.$$

Thus in both decompositions the simple groups that appear are, up to order,

$$\mathbb{Z}/2\mathbb{Z}, \quad \mathbb{Z}/2\mathbb{Z}, \quad \mathbb{Z}/3\mathbb{Z}.$$

1(f)

We must show that there are non-isomorphic groups with the same Jordan–Hölder decomposition.

A simple example is

$$\mathbb{Z}/4\mathbb{Z} \quad \text{and} \quad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}.$$

These groups are not isomorphic: the first is cyclic, while the second is not.

However, both groups have composition factors

$$\mathbb{Z}/2\mathbb{Z}, \quad \mathbb{Z}/2\mathbb{Z}.$$

Indeed, for $\mathbb{Z}/4\mathbb{Z}$ one has

$$\mathbb{Z}/4\mathbb{Z} \supset 2\mathbb{Z}/4\mathbb{Z} \supset \{0\},$$

while for $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ one may take any subgroup of order 2.

Exercise 2: Spontaneous symmetry breaking

We consider a particle on a circle. The Hilbert space is

$$\mathcal{H} = L^2(\mathbb{R}/\mathbb{Z}),$$

that is, square-integrable functions on the circle. Concretely, these are square-integrable functions $f(x)$ on $\mathbb{R}$ satisfying

$$f(x+1) = f(x).$$

The Hamiltonian is

$$H = \frac{1}{2}(-i\partial_x - \pi)^2.$$

We compute its eigenfunctions and study the symmetry

$$(Sf)(x) = e^{2\pi i x} f(-x).$$

Eigenfunctions of H

A natural orthonormal basis of $L^2(\mathbb{R}/\mathbb{Z})$ is given by the Fourier modes

$$f_n(x) = e^{2\pi i n x}, \quad n \in \mathbb{Z}.$$

These are periodic because

$$f_n(x+1) = e^{2\pi i n(x+1)} = e^{2\pi i n} e^{2\pi i n x} = f_n(x).$$

Now

$$\partial_x f_n = 2\pi i n f_n, \quad -i\partial_x f_n = 2\pi n f_n.$$

Hence

$$(-i\partial_x - \pi)f_n = (2\pi n - \pi)f_n = \pi(2n - 1)f_n.$$

Applying the square gives

$$Hf_n = \frac{1}{2}\pi^2(2n - 1)^2 f_n.$$

So the eigenfunctions are the Fourier modes f_n with eigenvalues

$$E_n = \frac{\pi^2}{2}(2n - 1)^2.$$

The symmetry operator

Define

$$(Sf)(x) = e^{2\pi ix} f(-x).$$

We first check periodicity:

$$(Sf)(x+1) = e^{2\pi i(x+1)} f(-(x+1)) = e^{2\pi ix} f(-x-1) = e^{2\pi ix} f(-x) = (Sf)(x),$$

using periodicity of f .

Now compute its action on the basis functions:

$$(Sf_n)(x) = e^{2\pi ix} f_n(-x) = e^{2\pi ix} e^{-2\pi inx} = e^{2\pi i(1-n)x} = f_{1-n}(x).$$

Thus

$$Sf_n = f_{1-n}.$$

Since

$$E_{1-n} = \frac{\pi^2}{2}(2(1-n)-1)^2 = \frac{\pi^2}{2}(2n-1)^2 = E_n,$$

S maps each eigenfunction to another eigenfunction with the same eigenvalue. So S is a symmetry of the Hamiltonian.

The ground state energy is obtained for $n = 0$ and $n = 1$, since

$$(2 \cdot 0 - 1)^2 = (2 \cdot 1 - 1)^2 = 1.$$

Thus the ground state is doubly degenerate, with eigenfunctions

$$f_0(x) = 1, \quad f_1(x) = e^{2\pi ix}.$$

The symmetry exchanges them:

$$Sf_0 = f_1, \quad Sf_1 = f_0.$$

Hence each individual ground state is not invariant under the symmetry, although the ground-state subspace is. This is spontaneous symmetry breaking in this example.

Exercise 3: Symmetries of the dodecahedron

We use the orbit–stabilizer theorem.

Definition. If a group G acts on a set X , the *orbit* of $x \in X$ is

$$G \cdot x = \{g \cdot x : g \in G\},$$

and the *stabilizer* of x is

$$G_x = \{g \in G : g \cdot x = x\}.$$

The orbit–stabilizer theorem states that

$$|G| = |G \cdot x| |G_x|.$$

3(a)

Let G be the symmetry group of the dodecahedron. We let G act on the set of its faces.

Since a dodecahedron has 12 faces, and any face can be mapped to any other by a symmetry, the action is transitive. Thus the orbit of a fixed face F has size

$$|G \cdot F| = 12.$$

By the orbit-stabilizer theorem,

$$|G| = |G \cdot F| \cdot |G_F| = 12 \cdot |G_F|.$$

Now G_F consists of all symmetries that fix the face F . Such symmetries act as symmetries of the regular pentagon forming that face. The symmetry group of a regular pentagon has 10 elements (5 rotations and 5 reflections), so

$$|G_F| = 10.$$

Therefore,

$$|G| = 12 \cdot 10 = 120.$$

3(b)

Let G be the full symmetry group of the dodecahedron, and let G_{rot} be the set of rotational symmetries.

From part (a), we know that

$$|G| = 120.$$

We now show that exactly half of these symmetries are rotations.

Choose one fixed orientation-reversing symmetry s of the dodecahedron, for example a reflection. Define a map

$$\Phi : G_{\text{rot}} \rightarrow G \setminus G_{\text{rot}}$$

by

$$\Phi(r) = sr.$$

Since r preserves orientation and s reverses orientation, the composition sr reverses orientation. Thus Φ indeed maps rotations to non-rotations.

We show that Φ is injective. If

$$\Phi(r_1) = \Phi(r_2),$$

then

$$sr_1 = sr_2.$$

Multiplying on the left by s^{-1} gives

$$r_1 = r_2.$$

So Φ is injective.

Now let $g \in G \setminus G_{\text{rot}}$ be any orientation-reversing symmetry. Then $s^{-1}g$ preserves orientation, because it is the composition of two orientation-reversing symmetries. Hence

$$r = s^{-1}g \in G_{\text{rot}}.$$

Therefore

$$\Phi(r) = s(s^{-1}g) = g.$$

So Φ is surjective.

Thus Φ is a bijection between the rotational symmetries and the orientation-reversing symmetries. Hence these two sets have the same size.

Since together they form the full symmetry group G , we conclude that

$$|G_{\text{rot}}| = \frac{|G|}{2} = \frac{120}{2} = 60.$$

Therefore the group of rotational symmetries of the dodecahedron has size

$$60.$$

3(c)

Inside the dodecahedron, there are five distinct cubes whose vertices are among the vertices of the dodecahedron. Let $C_1, \dots, C_5$ denote these cubes.

Every rotational symmetry of the dodecahedron permutes these five cubes. Thus we obtain a map

$$\rho : G_{\text{rot}} \rightarrow S_5,$$

defined by

$$\rho(g)(i) = j \quad \text{if } g(C_i) = C_j.$$

We first show that ρ is a homomorphism. For $g, h \in G_{\text{rot}}$ and any i ,

$$\rho(gh)(i) = \text{index of } (gh)(C_i) = g(h(C_i)) = \rho(g)(\rho(h)(i)).$$

Hence

$$\rho(gh) = \rho(g)\rho(h).$$

Next we show that ρ is injective. Suppose $\rho(g)$ is the identity permutation. Then $g(C_i) = C_i$ for all $i = 1, \dots, 5$, so g fixes each of the five cubes.

The vertices of these five cubes together are exactly the 20 vertices of the dodecahedron. Hence g fixes all vertices of the dodecahedron. A rotational symmetry fixing all vertices must be the identity. Therefore $g = 1$.

Thus $\ker(\rho) = \{1\}$, and ρ is injective.

3(d)

We show that the image of ρ is contained in the alternating group A_5 .

Definition. The alternating group A_5 is the subgroup of S_5 consisting of even permutations.

We show that every rotation induces an even permutation of the five cubes.

Rotational symmetries of the dodecahedron come in three types:

- Rotations of order 5 about an axis through opposite faces induce a 5-cycle on the cubes:

$$(C_1 C_2 C_3 C_4 C_5),$$

which is even since a 5-cycle is a product of 4 transpositions.

- Rotations of order 3 about an axis through opposite vertices induce a 3-cycle on three cubes and fix the other two:

$$(C_1 C_2 C_3),$$

which is even.

- Rotations of order 2 about an axis through midpoints of opposite edges induce a product of two disjoint transpositions:

$$(C_1 C_2)(C_3 C_4),$$

which is also even.

In all cases the induced permutation is even. Therefore

$$\rho(G_{\text{rot}}) \subseteq A_5.$$

Now ρ is injective and $|G_{\text{rot}}| = 60$. Since

$$|A_5| = \frac{5!}{2} = 60,$$

it follows that

$$\rho(G_{\text{rot}}) = A_5.$$

Hence

$$G_{\text{rot}} \cong A_5.$$

3(e)

We now describe the full symmetry group. We have already shown that the rotational subgroup is isomorphic to A_5 and has index 2. The extra symmetry is given by orientation reversal; this contributes a factor of $\mathbb{Z}_2$.

To make the splitting explicit, let ι denote the central inversion through the centre of the dodecahedron,

$$\iota(x) = -x.$$

This map has order 2 and commutes with every rotational symmetry, since

$$\iota(Rx) = -Rx = R(-x) = R\iota(x)$$

for every rotation R . Thus $\langle \iota \rangle \cong \mathbb{Z}_2$ is a central subgroup.

Every symmetry of the dodecahedron is either a rotation or a rotation composed with ι . Therefore every element can be written uniquely as

$$R\iota^\varepsilon, \quad R \in A_5, \varepsilon \in \{0, 1\}.$$

Since the $\mathbb{Z}_2$ -factor generated by ι commutes with the rotational subgroup, the product is direct. Hence

$$\text{Sym}(\text{dodecahedron}) \cong A_5 \times \mathbb{Z}_2.$$

Exercise 4: Cayley's theorem

Statement. Every finite group is isomorphic to a subgroup of a symmetric group.

Let G be a finite group. We let G act on itself by left multiplication:

$$g \cdot x = gx.$$

For each $g \in G$, define a map

$$L_g : G \rightarrow G, \quad L_g(x) = gx.$$

Since multiplication by g is bijective, L_g is a permutation of the set G . Thus each L_g is an element of the symmetric group S_G on the set G .

Now define

$$\Phi : G \rightarrow S_G, \quad \Phi(g) = L_g.$$

We check that this is a group homomorphism. For any $x \in G$,

$$\Phi(gh)(x) = L_{gh}(x) = (gh)x = g(hx) = L_g(L_h(x)) = (\Phi(g)\Phi(h))(x).$$

Hence

$$\Phi(gh) = \Phi(g)\Phi(h).$$

So Φ is a homomorphism.

To show that Φ is injective, suppose

$$\Phi(g) = \Phi(h).$$

Then $L_g = L_h$. Evaluating both sides at the identity element $1 \in G$, we get

$$g = L_g(1) = L_h(1) = h.$$

Thus Φ is injective.

Therefore G is isomorphic to the subgroup $\Phi(G) \subseteq S_G$. If $|G| = n$, then $S_G \cong S_n$, so every finite group embeds into some symmetric group. This is Cayley's theorem.